METRIC

A-A-59003/9
31 May 1995
Superseding
MIL-A-21577/9
31 January 1964

COMMERCIAL ITEM DESCRIPTION

AMPLIFIER-CONTROL GROUP AN/SIA-122

The General Services Administration has authorized the use of this commercial item description as a replacement for MIL-A-21577/9 which is canceled.

1. **Scope**. This commercial item description covers the detailed requirements for an amplifier voice intercommunication system for transmission of two-way communications between various aviation ordnance and missile handling stations in an aircraft carrier.

2. **Salient characteristics**.

2.1 **General**. Amplifier-control group AN/SIA-122(V) shall be in accordance with the applicable paragraphs of A-A-59003 and as specified herein. Whenever a requirement of A-A-59003 conflicts with a requirement of this specification, the requirement of this specification shall govern.

2.2 **Equipment**. The system complement shall consist of an amplifier-control assembly, two 30-station control units, and control, speaker-talk stations with explosion-proof loudspeakers. The quantity of speakers and control, speaker-talk stations shall be as specified in the contract or order. Requirements for the explosion proof loudspeakers is not covered in this specification.

2.3 **Amplifier-control assembly**. The amplifier-control assembly is the basic equipment in the system. The assembly shall provide for two independent amplified audio channels from which up to 30 speaker-talk-back stations per channel may be operated. The assembly shall also provide "audio override" by the speaker-talk-back stations.

2.3.1 **Enclosure**. The enclosure shall be formed sheet metal constructed with internal supporting frame work. The enclosure shall be designed for bulkhead mounting. The enclosure shall be drip-proof protected from the top. Forced air cooling shall not be permitted. Amplifier modules shall be mounted in a vertical position to facilitate the unrestricted flow of air.

2.3.2 **Maximum dimensions and weight**. The dimensions shall not exceed 407 mm in height, 356 mm in width, and 356 mm in depth. Weight shall not exceed 45.36 kg.

2.3.3 **Controls**. The following controls and components shall be provided on the front of the enclosure:

AMSC N/A

FSC 6320

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(a) Two 20-watt plug-in type power amplifiers with individual power supplies, one for each audio amplifier channel.

(b) Two amplifier power supply switches, one for each audio amplifier channel, connects ship power supply originating from 30-station control units to amplifiers.

(c) Two relay power supply switches, one for each audio channel, connects ships power supply originating from 30-station control units to relay power supplies.

(d) Two power available indicator lamps, indicates when the two power supplies are connected to amplifier control from the 30-station control units.

(e) One amplifier output test meter, indicates output level of amplifier channel under test.

(f) Two test switches, connects to test meter output and load resistor to channel under test.

(g) Two microphone receptacles, connects an MPM-6 microphone to amplifier for test purposes.

(h) Eight fuses for relay power and amplifier power supply protection.

Relays, wiring, terminals, and other components as required shall be provided within the enclosure.

2.3.4 **Relays**. Relays except power type contactors may be hermetically sealed.

2.3.5 **Alarm oscillator**. Alarm oscillation will not be required.

2.3.6 **Power amplifier modules**. - The power amplifier modules shall be constructed with quick disconnect features designed to permit easy removal of the modules from the amplifier control for servicing. The amplifier modules shall be provided with individual gain controls accessible through the front panel.

2.3.6.1 **Output power**. The rated output power for the individual power amplifier modules shall be 20 watts, minimum rating.

2.3.6.2 **Input impedance**. Input impedance for the individual power amplifier modules shall be 150 ohms (plus 10 minus 0 ohms) resistive at 1000 cycles per second (cps).

2.3.6.3 **Frequency response**. Frequency response of the power amplifiers shall be measured in accordance with A-A-59003, except that the amplifiers shall be delivering 10 decibels (db) below rated output at 1000 cps.

2.3.6.4 **Distortion**. Distortion of the power amplifiers shall be measured in accordance with A-A-59003, except that at 200 and 6000 cycles the measurement shall be made at 6 db below rated output.

2.3.6.5 **Semiconductor devices**. Semiconductor devices shall be employed.

2.3.6.6 **Interstage input circuit**. Input impedance and input level control will not be required.

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2.3.7 **Accessories.** A 6-foot test cable shall be provided for testing power amplifier modules when removed from the amplifier control.

2.4 **Thirty station control assembly.** The two 30-station control assemblies shall be identical in construction and operation and each assembly shall provide facilities for selecting and connecting any number of the 30 speaker-talk-back stations to one channel of the amplifier control assembly. Operation of any of the station selector switches, followed by operation of the speaker-talk-listen switch shall operate the necessary relays in the amplifier control assembly to provide audio communication to the selected speaker-talk-back stations. In addition to the audio communication, operation of the selector switch shall provide a busy indication at all 30 speaker-talk-back stations. Release of the speakers-talk-listen switch shall allow hands-free reply from speaker-talk-back station selected.

2.4.1 **Enclosure.** The enclosure shall be formed sheet metal designed for desk mounting. The enclosure shall be drip-proof from the top.

2.4.2 **Maximum dimensions and weight.** The dimensions shall not exceed 254 mm in height, 407 mm in width, and 204 mm in depth. Weight shall not exceed 11.34 kg.

2.4.3 **Controls.** The following controls shall be provided on the front of the enclosure:

(a) Power supply switch, connects ship's power supply to one channel of the amplifier control assembly.

(b) Power available indicator, indicates when ship's power supply is available.

(c) Thirty talk-back station selector switches, connect remote speaker-talk-back stations to amplifier input.

(d) Loudspeaker (reproducer microphone).

(e) Volume control, controls level of incoming audio from the talk-back station.

(f) Press-to-talk switch, connects loudspeaker to amplifier input when used as a microphone.

(g) Two fuses, protect amplifier control assembly power supply.

(h) Busy indicator, indicates when system is in use.

2.4.4 **Loudspeaker.** The loudspeakers shall be type LS-445.

2.5 **Control, speaker-talk-station.** The control unit shall provide facilities for initiating emergency reports to its respective 30-station control unit. The "Emergency-Report" switch shall be provided speech priority over the 30-station control unit. In addition, operation of this switch shall provide busy indications at all 30 speaker-talk-back stations and the 30- station control unit.

2.5.1 **Enclosure.** The enclosure shall be designed for bulkhead mounting and shall be explosion proof and drip-proof.

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2.5.2 **Maximum dimensions and weight**. The dimensions shall not exceed 153 mm in height, 153 mm in width, and 127 mm in depth. Weight shall not exceed 4.54 kg.

2.5.3 **Controls**. The following controls shall be provided on the front of the enclosure:

(a) Busy indicator lamp, indicates when system is in use.

(b) Emergency report switch, allows operator to report and override outgoing reports from the 30-station control unit.

3. **Quality assurance**. Sampling, examination, and tests shall be in accordance with A-A-59003.

4. **Packaging**. Packaging shall be in accordance with A-A-59003.

5 **Notes.**

5.1 **Intended use**. Amplifier-control group AN/SIA-122(V) is intended for use in communication systems between aviation ordinance and missile handling stations on aircraft carriers.

5.2 **Notes**. The notes specified in A-A-59003 are applicable to this specification.

Preparing activity:
Navy - SH
(Project 6320-N055)

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